

Tutorial Sheet - 05

• Sec-A

1. Define functional dependencies.

Ans Functional dependencies in a relational database represent the relationship between attributes in a table, where the value of one attribute uniquely determines the value of another attribute.

2. Define the Super key, candidate key and Primary Key.

Ans Super Key: A super key is a set of one or more attributes that, taken collectively, uniquely identifies each tuple (row) in a relation (table). or it is a set of attributes that can uniquely identify each record in a table.

• Candidate key: A candidate key is a minimal super key. It's a super key that does not have any unnecessary attributes. It's a set of attributes that can uniquely identify each row in the table, but without any redundancy. A table can have multiple candidate keys.

• Primary key: A Primary key is a unique identifier for each record in a database table. The Primary key must be unique for each record and cannot contain null values.

3. What is the condition of transitive dependency in 3NF?

Ans

in 3NF, Condition for transitive dependency is that there should be no nonprime attribute that is transitively dependent on candidate key.

4. Define the types of anomalies in normalization.

Ans Normalization aims to reduce data anomalies by organizing data into well-structured relations. There are three types of anomalies

- (i) Insertion Anomaly: Difficulty in adding data due to missing required attributes.
- (ii) Update Anomaly: Inconsistency arising from updating data in one place but not in others.
- (iii) Deletion Anomaly: ~~an~~ occurs when updating data in one place requires updating it in multiple places to ensure consistency, which is error-prone and inefficient.

5. What are redundant variables? For the following relation $R(A, B, C, D, E)$ and functional dependencies: $A \rightarrow B$, $C \rightarrow A$, $AB \rightarrow D$. Find the prime and nonprime attributes.

Ans Redundant variables are those variable that are duplicate or highly correlated with others.

$R(A, B, C, D, E)$

$CE^+ = CEABD$

So CE is a candidate key

Prime attribute = C, E

Non-Prime = A, B, D

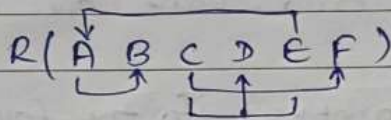
• Sec-B ($3 \times 7 = 21$)

1. How to find functional dependency for a relation?
Let $R(A, B, C, D, E, F)$ be a relation schema with the following functional dependencies $C \rightarrow F$, $E \rightarrow A$, $EC \rightarrow D$, $A \rightarrow B$, which is the C.K for R and find out the closure for all.

Ans To find F.D in relation, we analyze data and look for pattern and relationship between attributes. We examine values in relation and determine how one attribute depends on another.

Step -

- Identify attributes
- Examine the data
- Use inference rules
- test for dependency



F.D:

$$C \rightarrow F$$

$$E \rightarrow A$$

$$EC \rightarrow D$$

$$A \rightarrow B$$

$$CE^+ = CEAFDB$$

So

CE is a Candidate key.

Now find closure of each:

$$A^+ = \{AB\}$$

$$B^+ = \{B\}$$

$$C^+ = \{CF\}$$

$$D^+ = \{D\}$$

$$E^+ = \{EAB\}$$

$$F^+ = \{F\}$$

Q2. What is the Closure Method? Suppose, a relational Schema $R(A, B, C, D, E, F)$ and set of F.D.:

$$F \{ AB \rightarrow C, BC \rightarrow AD, D \rightarrow E, CF \rightarrow B \}$$

Compute BCF^+ , CD^+ , D^+ .

Ans. Closure Method also known as Armstrong's axioms or closure properties is a technique in DBMS to determine complete set of F.D. for given set of attributes. It helps in identifying all possible dependencies that exist in a relation.

$R(A, B, C, D, E, F)$

F.D.:

$AB \rightarrow C$

$BC \rightarrow AD$

$D \rightarrow E$

$CF \rightarrow B$

$BCF^+ = \{ B, C, F, A, D, E \}$

$CD^+ = \{ C, D, E \}$

$D^+ = \{ D, E \}$

Q3. A Relation $R(ABCD)$ having two F.D. sets P
 $FD1 = \{ A \rightarrow B, B \rightarrow C, AB \rightarrow D \}$ and
 $FD2 = \{ A \rightarrow B, B \rightarrow C, A \rightarrow C, A \rightarrow D \}$. Find the equivalence of F.D. How can we conclude that two F.D. are equivalent?

Ans.

Let's say $FD1$ as F
 and $FD2$ as G

F	G
$A \rightarrow B$	$A \rightarrow B$
$B \rightarrow C$	$B \rightarrow C$
$AB \rightarrow D$	$A \rightarrow C$
	$A \rightarrow D$

• Closure of F using F

$$A^+ = \{ABCD\}$$

$$B^+ = \{BC\}$$

$$AB^+ = \{ABCD\}$$

Closure of F using G

$$A^+ = \{ABCD\}$$

$$B^+ = \{BC\}$$

$$AB^+ = \{ABCD\}$$

So, $F \subseteq G$

• Closure of G using G

$$A^+ = \{ABCD\}$$

$$B^+ = \{BC\}$$

Closure of G using F

$$A^+ = \{ABCD\}$$

$$B^+ = \{BC\}$$

So, $G \subseteq F$

As, $F \subseteq G$ and $G \subseteq F$ So $F = G$

So given two F.D. are equivalent.

In general, to conclude that two sets of F.D. are equivalent, we need to demonstrate that each F.D. in one set can be derived from the other set using the Armstrong's axioms (Reflexivity, Augmentation, Transitivity).

• Sec-C (SX11233)

Q. Define Minimal Cover. Write the steps to find the canonical form of F.D. Find the minimal cover of the set of functional dependencies given;

$\{A \rightarrow C, AB \rightarrow C, C \rightarrow DI, CD \rightarrow I, EC \rightarrow AB, EI \rightarrow C\}$

Ans. Minimal Cover is concise representation of a set of F.D that eliminate any redundant or extraneous dependencies. It is a way to simplify and optimize set of F.D while preserving same closure.

Steps -

- Start with given set of F.D.
- Eliminate redundant dependencies.
- Eliminate extraneous attributes.
- Eliminate transitive dependencies.
- Repeat steps 2-4 until no further change can be made.

F.D. ~~$A \rightarrow C$~~
 $AB \rightarrow C$
 $C \rightarrow DI$
 $CD \rightarrow I$
 $EC \rightarrow AB$
 $EI \rightarrow C$

Decompose:

$A \rightarrow C$	— (i) ✓
$AB \rightarrow C$	— (ii) ✓
$C \rightarrow D$	— (iii) ✓
$C \rightarrow I$	— (iv) X
$CD \rightarrow I$	— (v) ✓
$EC \rightarrow A$	— (vi) ✓
$EC \rightarrow B$	— (vii) ✓
$EI \rightarrow C$	— (viii) ✓

$$A^+ = \{ACDI\}$$

$$A^+ \text{ without } \textcircled{I} = \{A\}$$

both are different so it is essential.

$$AB^+ = \{ABCDI\}$$

$$AB^+ \text{ without } \textcircled{II} = \{AB\}, \text{ essential.}$$

$$C^+ = \{CDI\}$$

$$C^+ \text{ without } \textcircled{III} = \{CI\} \text{ essential}$$

$$C^+ \text{ without } \textcircled{IV} = \{CDI\} \rightarrow \text{Not essential.}$$

$$CD^+ = \{CDI\}$$

$$CD^+ \text{ without } \textcircled{V} = \{CD\} \text{ essential}$$

$$EC^+ = \{CEABDI\}$$

$$EC^+ \text{ without } \textcircled{VI} = \{ECB\} \text{ essential}$$

$$EC^+ \text{ without } \textcircled{VII} = \{ECADI\} \text{ essential}$$

$$EI^+ = \{EICDBA\}$$

$$EI^+ \text{ without } \textcircled{VIII} = \{EI\} \text{ essential}$$

$$A \rightarrow C, AB \rightarrow C, C \rightarrow D, CD \rightarrow I, EC \rightarrow AB, EI \rightarrow C$$

Now,

Check for $AB \rightarrow C$

$$AB^+ = \{ABCDI\}$$

$$A^+ = \{ACDI\}$$

$$B^+ = \{B\}$$

So $AB \rightarrow C$ is essential.

$$\rightarrow C \rightarrow I$$

$$C^+ = \{C, I\}$$

$$C^+ = \{C, I\}$$

$$D^+ = \{D\}$$

$$C \rightarrow I$$

So, here D is not ~~essential~~ required.

$\rightarrow$

$$EC \rightarrow AB$$

$$EC^+ = \{E, C, D, I, A, B\}$$

$$E^+ = \{E\}$$

$$C^+ = \{C, D, I\}$$

So, $EC \Rightarrow AB$ is essential.

$$\rightarrow EI \rightarrow C$$

$$EI^+ = \{E, I, C, D, A, B\}$$

$$E^+ = \{E\}$$

$$I^+ = \{I\}$$

So, $EI \rightarrow C$ is essential.

Finally - $A \rightarrow C, AB \rightarrow C, C \rightarrow D, C \rightarrow I, EC \rightarrow AB, EI \rightarrow C$.

or

M.C.:- $A \rightarrow C, AB \rightarrow C, C \rightarrow D, C \rightarrow I, EC \rightarrow AB, EI \rightarrow C$.

Q2. What is normalization? What is the purpose of normalization in DBMS? Given a relation $R(P, B, R, S, T)$ and F.D set $FD = \{P \rightarrow B, S \rightarrow T\}$, determine whether the given R is in 2NF? if not convert it into 2NF.

Ans Normalization is the process of organizing the data in the database. It is used to minimize the redundancy from a relation or set of relations.

Purpose:

- (i) Minimize ^{data} Redundancy.
- (ii) Enhance data integrity.
- (iii) Improve database performance.
- (iv) Prevent data update anomalies.
- (v) Facilitate Query optimization.

$R(P, O, R, S, T)$

F.D: $PO \rightarrow R$
 $S \rightarrow T$

$PO^{\#} = POSRT$

So C.K = POS

$PO \rightarrow R$ → Partial Dependency
 ↓ ↓
 Prime Non-Prime

So NO in 2NF.

$S \rightarrow T$
 ↓ ↓
 Prime non-prime

$R_1(POR)$

$R_2(ST)$

$R_3(POS)$

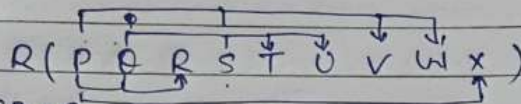
R_1, R_2, R_3 are in 2NF.

Q3. Explain BCNF. Given a relation $R(PORSTUVWX)$ and F.D set $F.D = \{PO \rightarrow R, OS \rightarrow TU, PS \rightarrow VW, \text{ and } P \rightarrow X\}$, determine whether the given R is in which normal form?

Ans. BCNF stands for Boyce-Codd Normal Form, which is a higher level of third normal form (3NF) and aims to further eliminate redundancy and anomalies in

a database schema.

A relation is in BCNF if, for every one of its non-trivial functional dependencies ($x \rightarrow y$), x is a superkey.



F.D: $PQ \rightarrow R$

$QS \rightarrow TU$

$PS \rightarrow VW$

$P \rightarrow X$

So $PQS^+ = PQSRTUVWX$

$PQS =$ candidate key.

$(PQ) \rightarrow R$

✓

prime

PQ is not a super key neither R is prime attribute so this is not in BCNF.
and not in 3NF.

Let check for 2NF.

$PQ \rightarrow R$

↓
prime

↓
non-prime

Here PQ depend on R which is a non-prime attributes

So here partial dependency exist.

Hence it is not in 2NF.

So, from the above three statements, we can say that table $R(PQSTUVWX)$ is in 1NF only.